

100, 90, 80, 70, 60, 50, 40, 30, 20, 10.

start stop

$i = 100$

$i \geq 10$

$i = 100$

$100 \geq 10$

$i = 90$

$90 \geq 10$

$i = 80$

$80 \geq 10$

$i = 70$

$70 \geq 10$

$i = 60$

$60 \geq 10$

$i = 50$

$50 \geq 10$

$i = 40$

$40 \geq 10$

$i = 30$

$30 \geq 10$

$i = 20$

$20 \geq 10$

$i = 10$

$10 \geq 10$

$i = 0$

$0 \geq 10$

(False)

True

40

30

20

10

0

step

$i = i - 10$

$i = 100 - 10$

$\Rightarrow 90$

$i = 90 - 10$

$\Rightarrow 80$

$i = 80 - 10$

$\Rightarrow 70$

$i = 70 - 10$

$\Rightarrow 60$

$i = 60 - 10 \Rightarrow 50$

$i = 50 - 10 \Rightarrow 40$

$i = 40 - 10 \Rightarrow 30$

$i = 30 - 10 \Rightarrow 20$

$i = 20 - 10 \Rightarrow 10$

$i = 10 - 10 \Rightarrow 0$

2) Sum of first 5 natural numbers.

1 2 3 4 5

$n = 5$

$sum = 0$

start

(i)

stop

$i \leq 5$

logic/code

Print

stop

$i = i + 1$

$i = 1$

$i \leq 5$

$sum = 0 + 1$   
 $= 1$

$1 + 1 = 2$

$i = 2$

$2 \leq 5$

$sum = 1 + 2$   
 $= 3$

$2 + 1 = 3$

$i = 3$

$3 \leq 5$

$sum = 3 + 3$   
 $= 6$

$3 + 1 = 4$

$i = 4$

$4 \leq 5$

$sum = 6 + 4$   
 $= 10$

$4 + 1 = 5$

$i = 5$

$5 \leq 5$

$sum = 10 + 5$   
 $= 15$

$5 + 1 = 6$

Ex find average of first n natural numbers  
using loop.

Ex  
(Start) if stop i val.

$$i = 1 \text{ loop } 1 < 6$$

$$i = 2 \quad 2 < 6$$

$$i = 3 \quad 3 < 6$$

$$i = 4 \quad 4 < 6$$

$$i = 5 \quad 5 < 6$$

$$i = 6 \quad 6 < 6 \text{ (False)}$$

code logic

stop

$$\text{sum} = 0 + 1 + 1 = 2$$

$$\text{sum} = 1 + 2 = 3$$

$$\text{sum} = 3 + 3 = 6$$

$$\text{sum} = 6 + 4 = 10$$

$$\text{sum} = 10 + 5 = 15$$

$$\text{sum} = 15$$

1. find avg of first n natural number using loop  
n=5

start - stop  
~~i=1~~ i < 6

i=2 2 < 6

i=3 3 < 6

i=4 4 < 6

i=5 5 < 6

i=6 6 < 6  
(X)

logic

$$\text{sum} = 0 + 1 = 1$$

$$\text{sum} = 1 + 2 = 3$$

$$\text{sum} = 3 + 3 = 6$$

$$\text{sum} = 6 + 4 = 10$$

$$\text{sum} = 10 + 5 = 15$$

step

$$i = 1 + 1 = 2$$

$$i = 2 + 1 = 3$$

$$i = 3 + 1 = 4$$

$$i = 4 + 1 = 5$$

$$i = 5 + 1 = 6$$

$$i = 5 + 1 = 6$$

$$\text{Avg} = \text{sum} / n = 15 / 5 = 3.0$$

2. sum of squares of numbers 1 to N.

N=2, sum=0

start

i=1

i=1

i=2

i=3

stop

i < n+1

1 < 3

2 < 3

3 < 3  
(X)

sum of square = 5

logic

$$\text{sum} = \text{sum} + i^2$$

$$\text{sum} = 0 + 1^2 = 1$$

$$\text{sum} = 1 + 2^2 = 5$$

step

$$i = i + 1$$

$$i = 1 + 1 = 2$$

$$i = 2 + 1 = 3$$

3. sum of cubes of number from 1 to N

n=3, sum=0.

start

i=1

i=1

stop

i < n+1

1 < 4

logic

$$\text{sum} = \text{sum} + i^3$$

$$\text{sum} = 0 + 1^3 = 1$$

step

$$i = i + 1$$

$$i = 1 + 1 = 2$$

$i=2$   $2 < 4$   $sum = 1 + 2^{**}3 = 9$   $i = 2+1 = 3$   
 $i=3$   $3 < 4$   $sum = 9 + 3^{**}3 = 26$   $i = 3+1 = 4$   
 $i=4$   $4 < 4$  (X)

sum of cube = 36.

4. Calculate Power of a number without using  $**$  operator.

base = 2, Power = 3, result = 1

Start  
 $i=1$

stop  
~~Power~~  
 $i < Power + 1$

Logic  
result = result \* base  $i = i + 1$

$i=1$

$1 < 4$

result =  $1 * 2 = 2$

$i = 1 + 1 = 2$

$i=2$

$2 < 4$

result =  $2 * 2 = 4$

$i = 2 + 1 = 3$

$i=3$

$3 < 4$

result =  $4 * 2 = 8$

$i = 3 + 1 = 4$

$i=4$

$4 < 4$  (X)

5. Display Fibonacci Series.

$n=5, a=0, b=1$

start

stop =

Logic

step.  $i=i$

$i=1$

$i < n+1$

$c = a + b$

$i = i + 1$

$a = b$

$b = c$

$i=2$

$2 < 6$

$a=0$   
 $c = 0 + 1$

$i = 1 + 1 = 2$

$a = 1$

$b = 1$

$i=3$

$3 < 6$

$a=1$   
 $c = 1 + 1 = 2$   
 $a = 1$   
 $b = 2$

$i = 2 + 1 = 3$

$$a=1$$

$$3 < 6$$

$$c = 1+2=3 \quad s = 3+1=4$$

$$a=2$$

$$b=3$$

$$c = 2+3=5 \quad s = 4+1=5$$

$$a=3$$

$$b=5$$

$$a=3$$

$$a=3$$

$$c = 3+5=8$$

$$a=5$$

$$b=7$$

$$a=5$$

$$r=5$$

$$5 < 6$$

$$6 < 6 \quad (X)$$

$$r=6$$

6. Display Series = 1, 1/2, 1/3, 1/4, ...

$$n=4$$

Start

$$i=1$$

stop

$$i < n+1$$

logic

$$if(1/i == 0)$$

$$print(i)$$

else

$$print(1/i)$$

step

$$i = i+1$$

$$i=1$$

$$1 < 5$$

$$1 \cdot 1 = 1 \neq 0 \quad i = 1+1 = 2$$

$$i=2$$

$$2 < 5$$

$$2 \cdot 1 = 2 \neq 0$$

$$i = 2+1 = 3$$

$$i=3$$

$$3 < 5$$

$$3 \cdot 1 = 3 \neq 0$$

$$i = 3+1 = 4$$

$$i=4$$

$$4 < 5$$

$$4 \cdot 1 = 4 \neq 0$$

$$i = 4+1 = 5$$

$$i=5$$

$$5 < 5$$

(X)

2. Display first N terms of Series.

Ex: 1, 11, 111, 1111, 11111

$n = 5$ , Series = ""

| Start   | Stop        | Logic                          | Step            |
|---------|-------------|--------------------------------|-----------------|
| $i = 1$ | $i < n + 1$ | Series = Series + "1"          | $i = i + 1$     |
| $i = 1$ | $1 < 5$     | Series = "" + "1"<br>= 1       | $i = 1 + 1 = 2$ |
| $i = 2$ | $2 < 5$     | Series = "1" + "1"<br>= 11     | $i = 2 + 1 = 3$ |
| $i = 3$ | $3 < 5$     | Series = "11" + "1"<br>= 111   | $i = 3 + 1 = 4$ |
| $i = 4$ | $4 < 5$     | Series = "111" + "1"<br>= 1111 | $i = 4 + 1 = 5$ |
| $i = 5$ | $5 < 5$ (X) |                                |                 |

3. Display N Series, each term obtained by multiply by 3 with Previous.

Ex: 1, 3, 9, 27, 81

~~n~~  $n = 5$ ,  $a = 1$

| Start   | Stop        | Logic                   | Step            |
|---------|-------------|-------------------------|-----------------|
| $i = 1$ | $i < n + 1$ | $a = a * 3$             | $i = i + 1$     |
| $i = 1$ | $1 < 5$     | $a = 1 * 3$<br>$a = 3$  | $i = 1 + 1 = 2$ |
| $i = 2$ | $2 < 5$     | $a = 3 * 3$<br>$a = 9$  | $i = 2 + 1 = 3$ |
| $i = 3$ | $3 < 5$     | $a = 9 * 3$<br>$a = 27$ | $i = 3 + 1 = 4$ |